Tema 3. Ejercicios de visualización

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Índice

1. Superposición y ajustes de posición **1**

2. Taller: CPI e índice de desarrollo humano **4**

3. Líneas de ajuste **8**

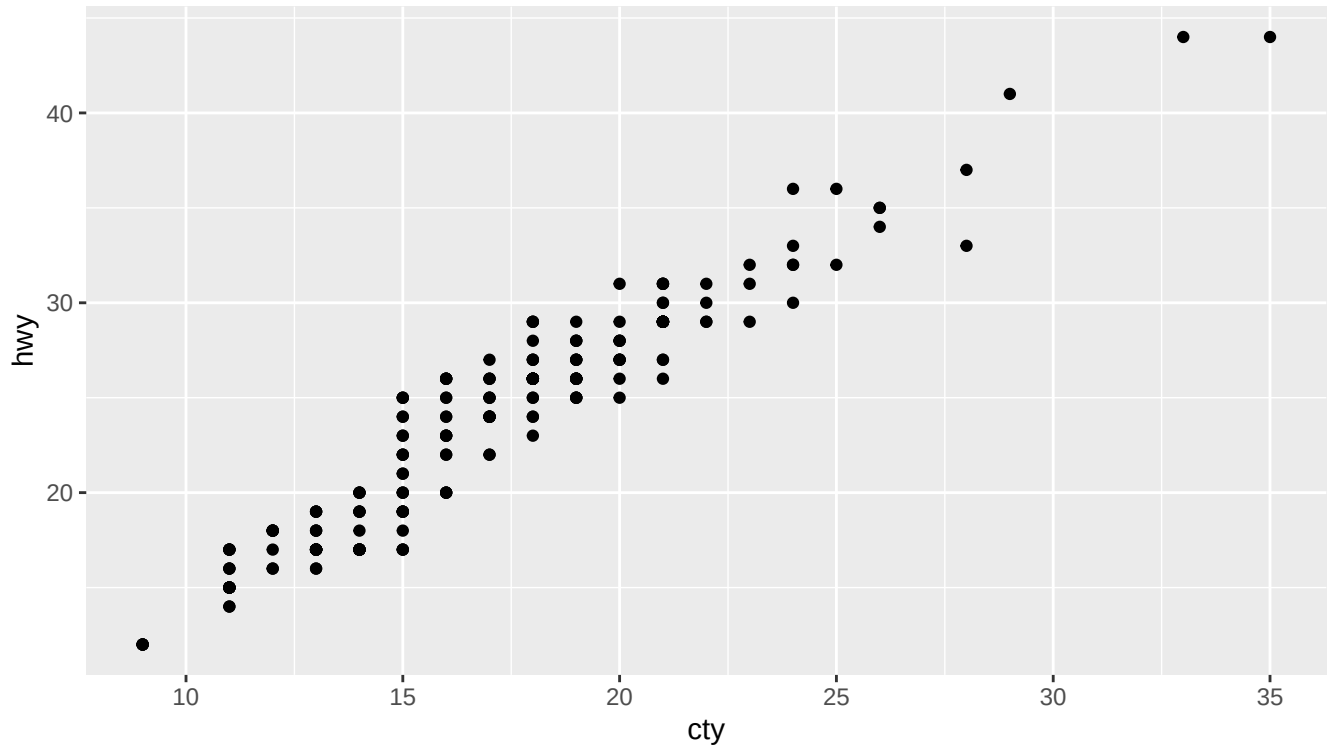
4. Etiquetas y colores **10**

5. Gráfico de The Economist **12**

1. Superposición y ajustes de posición

Hay coches con los mismos valores de consumo: los puntos se superponen. `geom_jitter()` los desplaza ligeramente; `geom_count()` cambia su tamaño según cuántas observaciones coinciden.

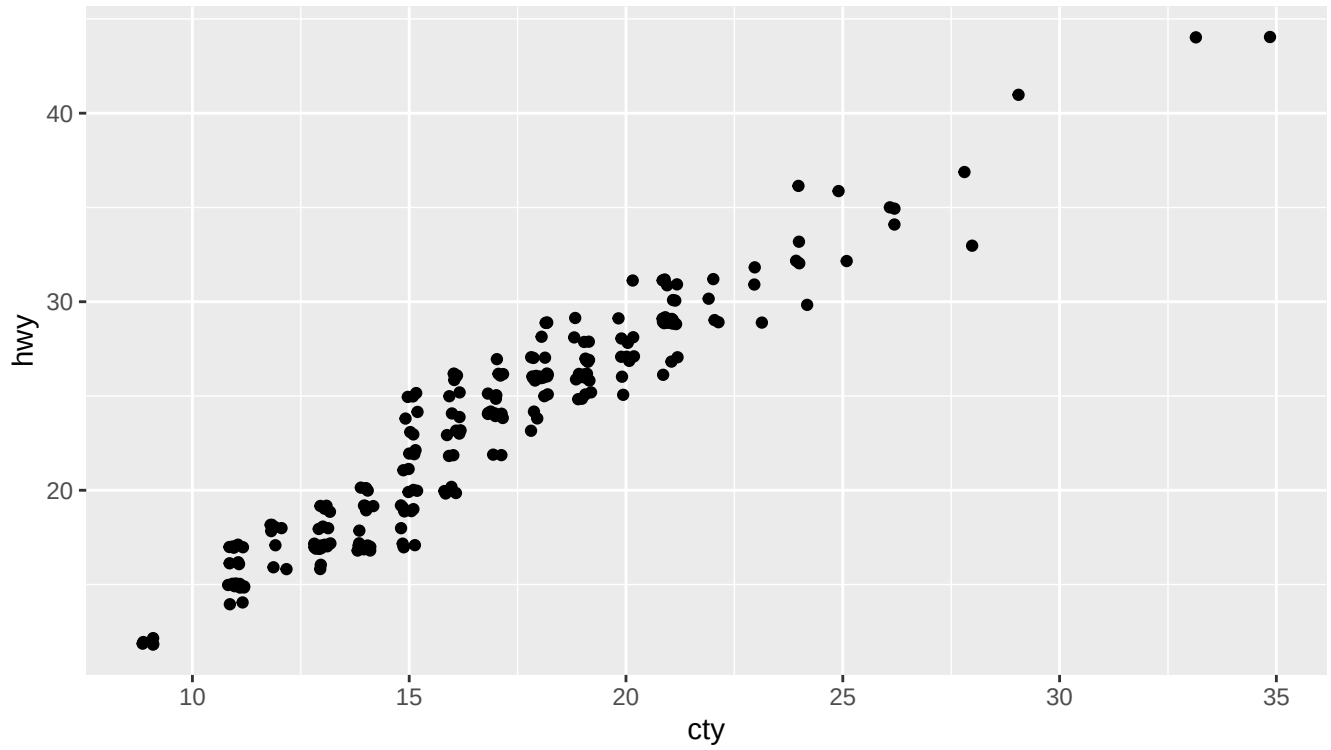
```
p <- ggplot(mpg, aes(x = cty, y = hwy))
p + geom_point()
```



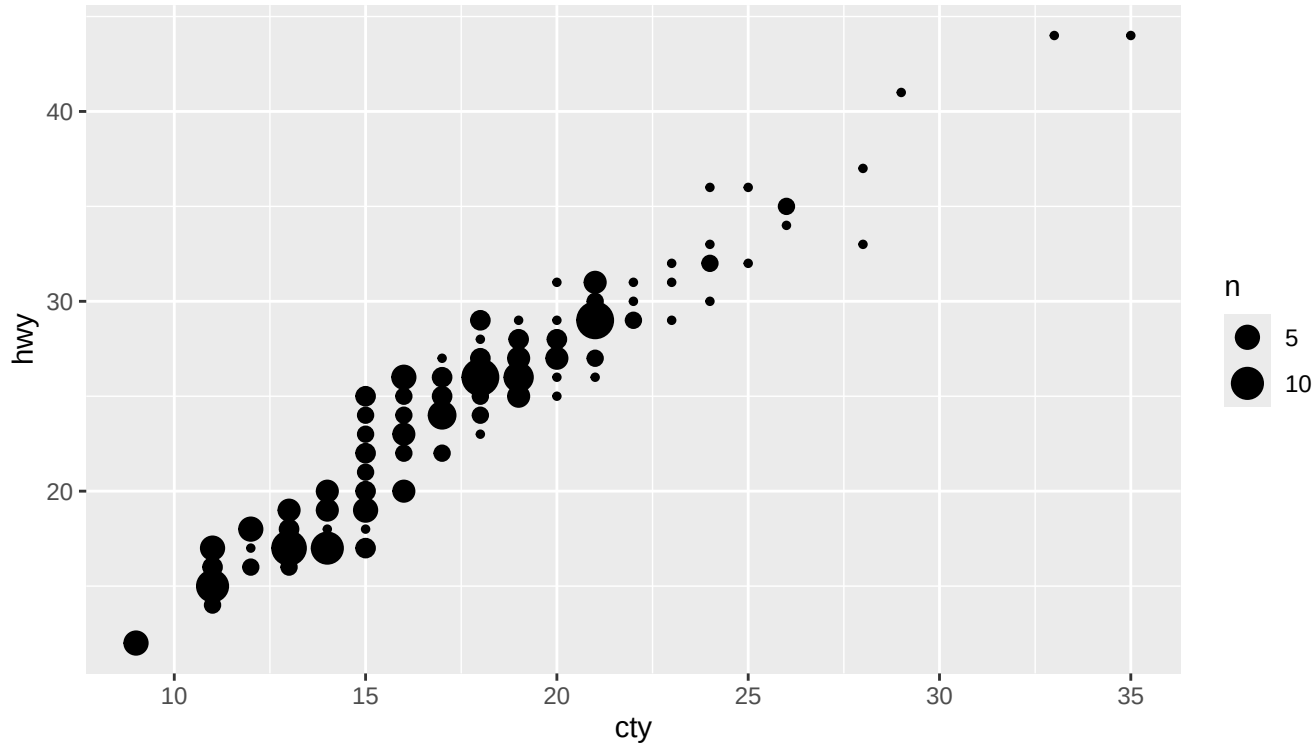

```
nrow(mpg) - nrow(distinct(mpg, cty, hwy))
```

```
[1] 156
```

```
p + geom_jitter(width = 0.2, height = 0.2)
```

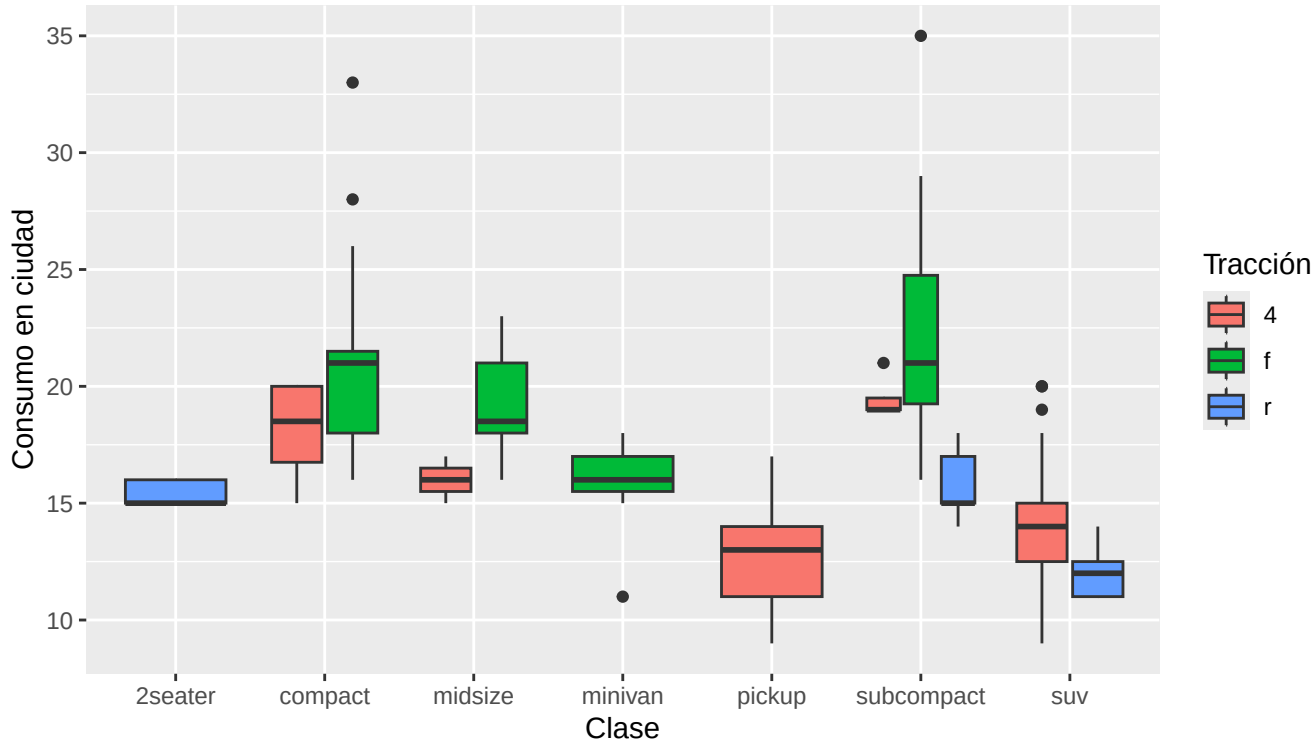



```
p + geom_count()
```



La posición predeterminada de `geom_boxplot()` es `dodge2`.

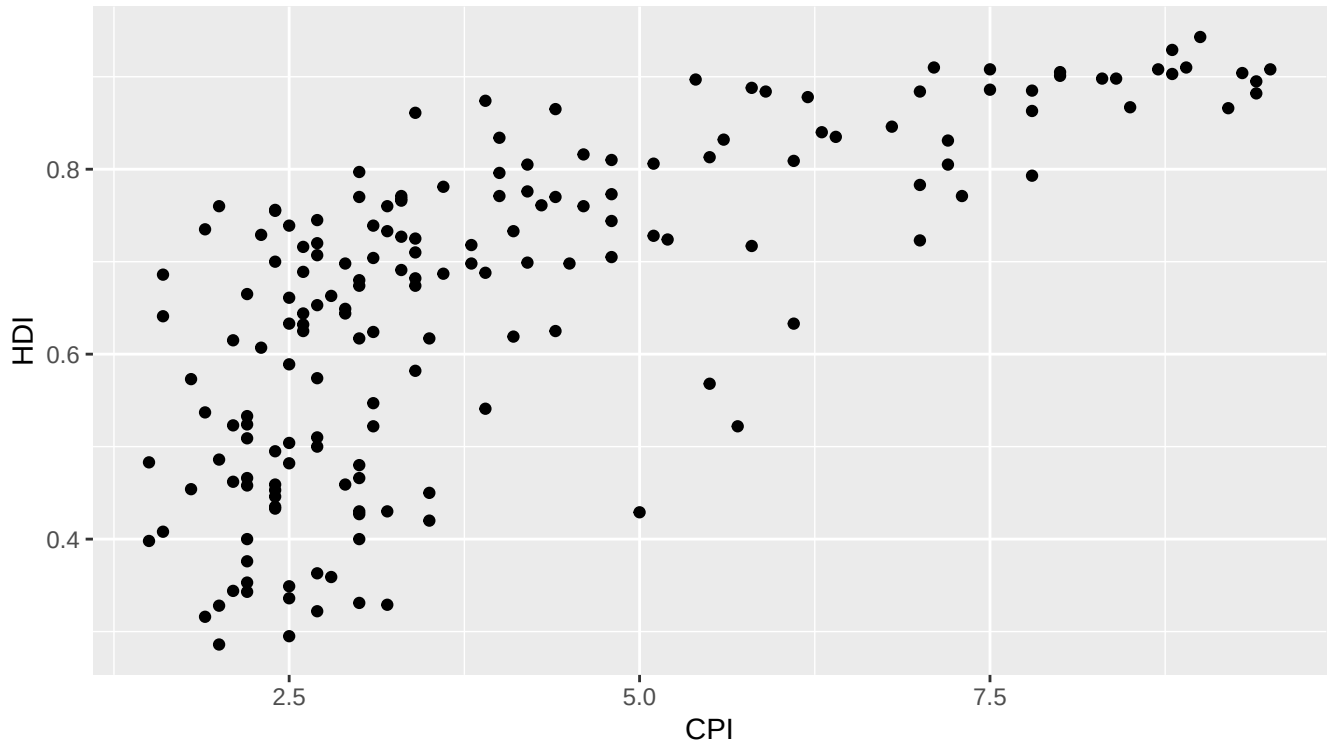
```
ggplot(mpg, aes(x = class, y = cty, fill = drv)) +
  geom_boxplot() + labs(x = 'Clase', y = 'Consumo en ciudad', fill = 'Tracción')
```



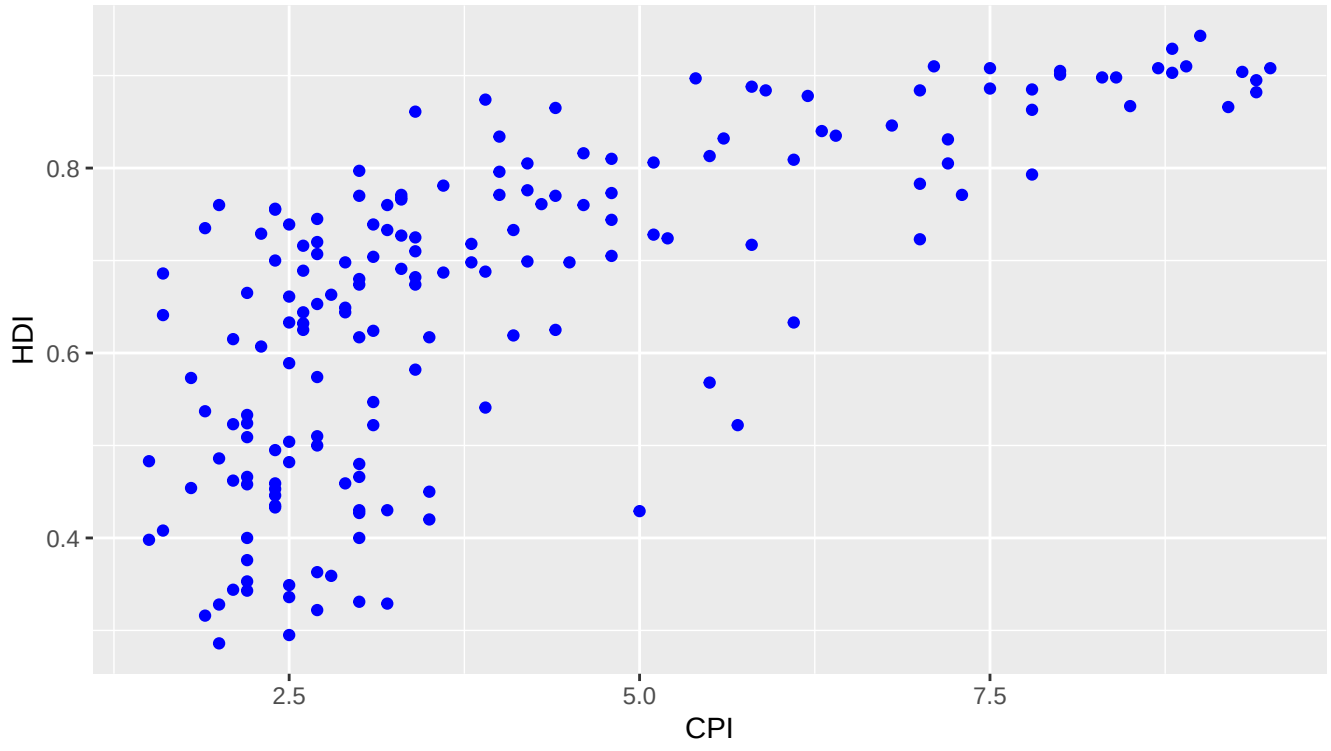
2. Taller: CPI e índice de desarrollo humano

Se cargaron los datos del taller y se representaron los indicadores.

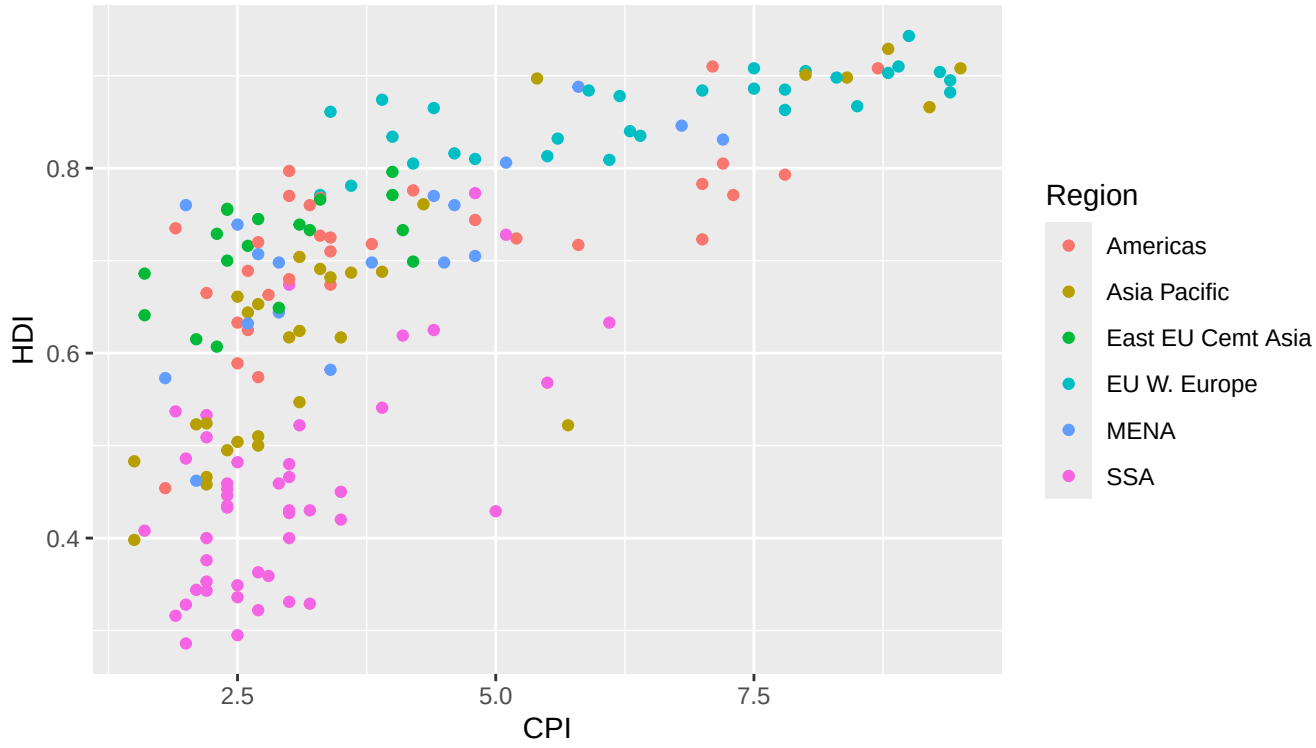
```
dat <- read.csv('data/EconomistData.csv')
p <- ggplot(dat, aes(x = CPI, y = HDI))
p + geom_point()
```




```
p + geom_point(colour = 'blue')
```

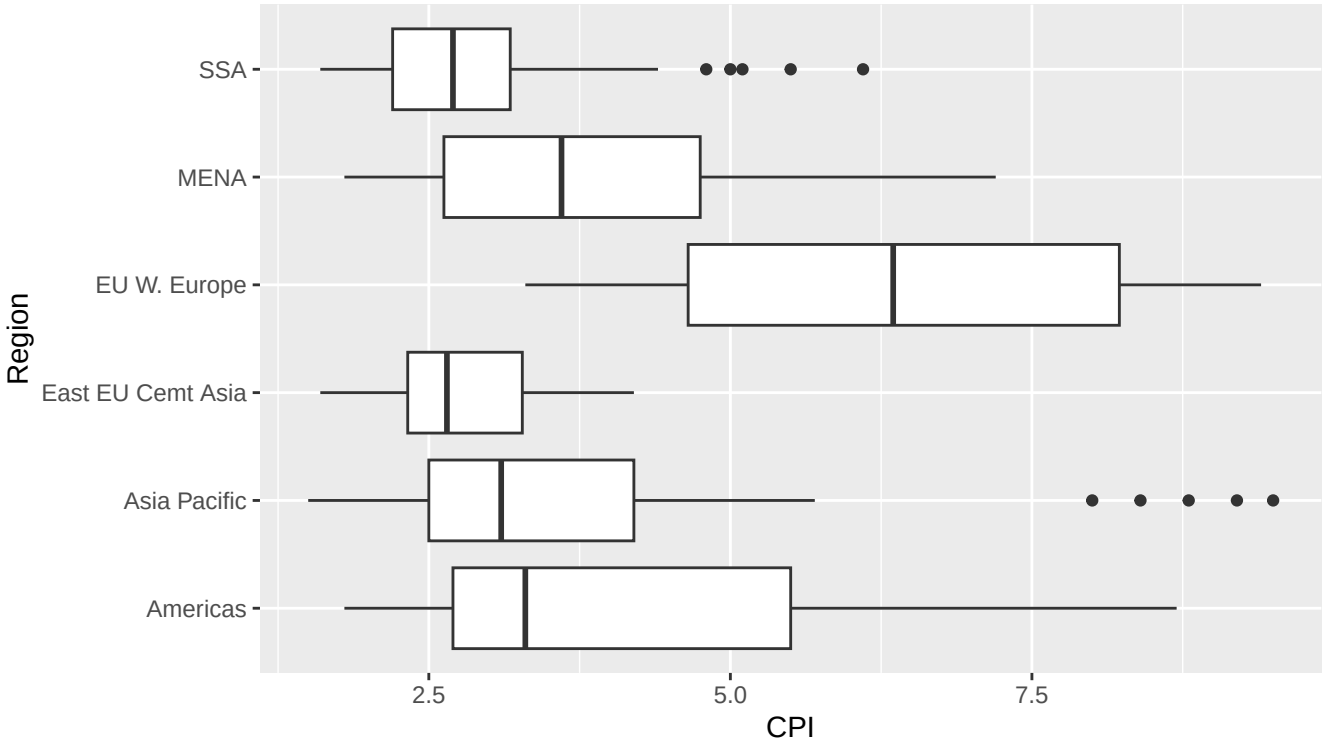


```
p + geom_point(aes(colour = Region))
```

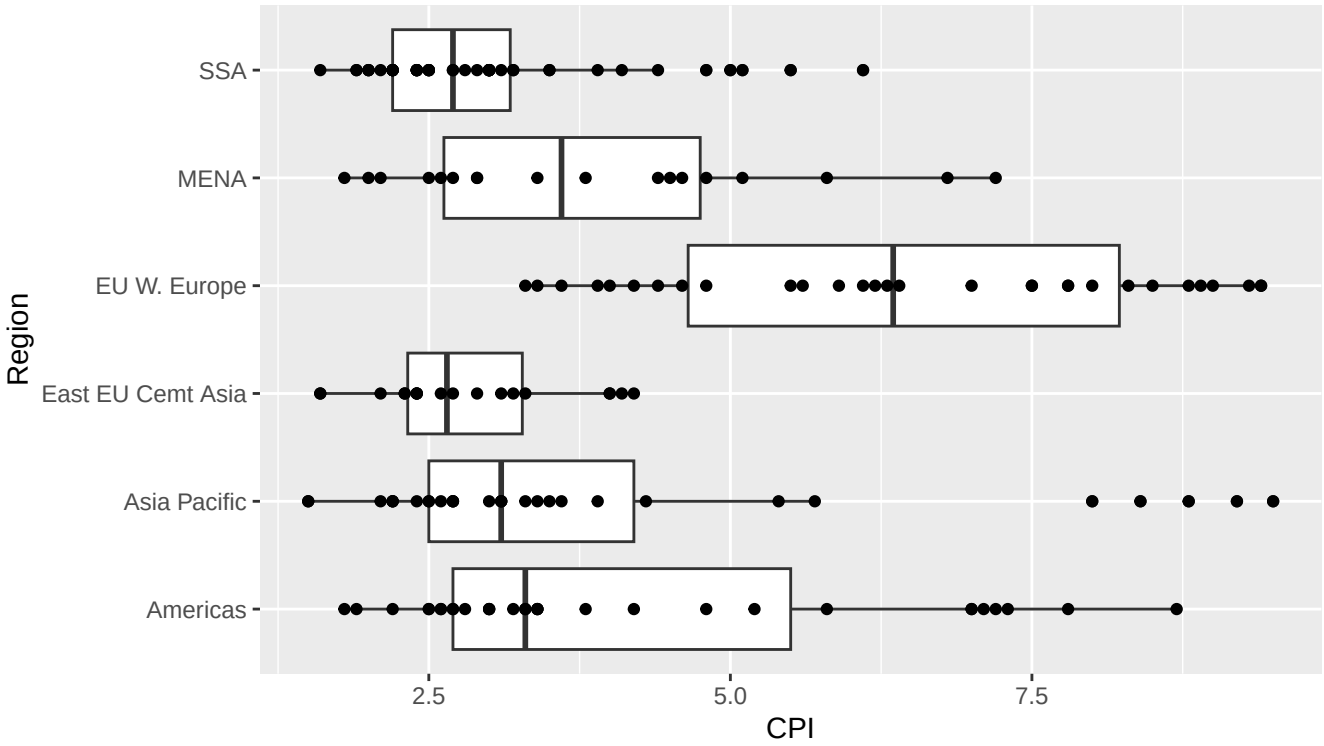


Los boxplots muestran el CPI por región; después se añadieron los puntos.

```
cajas <- ggplot(dat, aes(x = Region, y = CPI)) +
  geom_boxplot() + coord_flip()
cajas
```



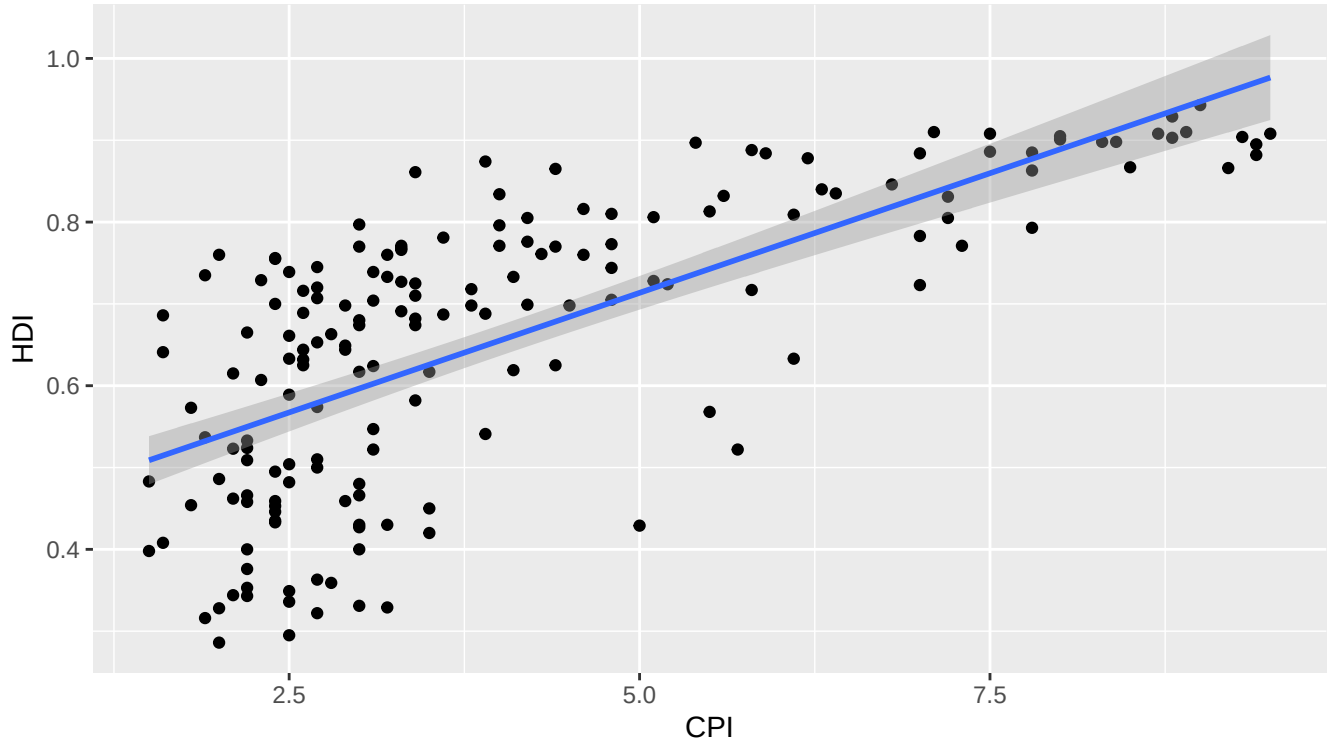

```
cajas + geom_point()
```



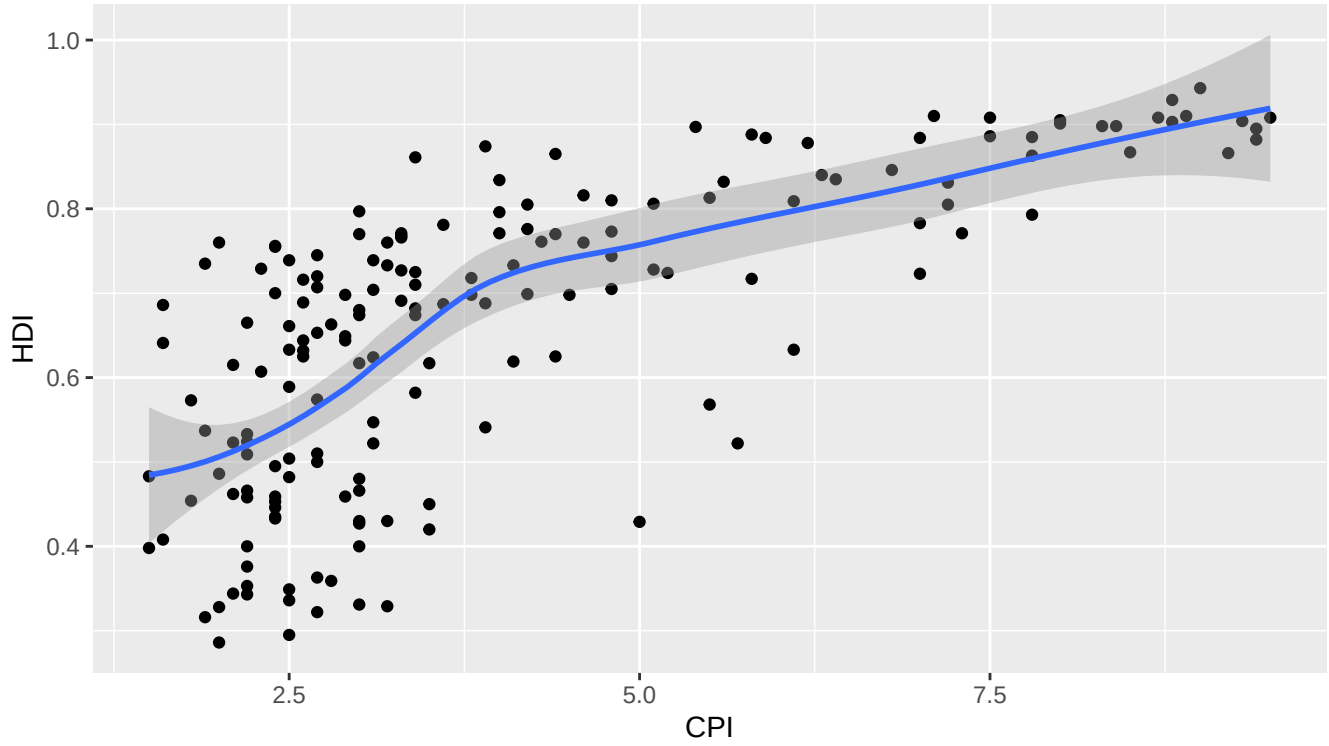
3. Líneas de ajuste

Se compararon los ajustes lineal y LOESS. Un `span` menor da una curva menos suave.

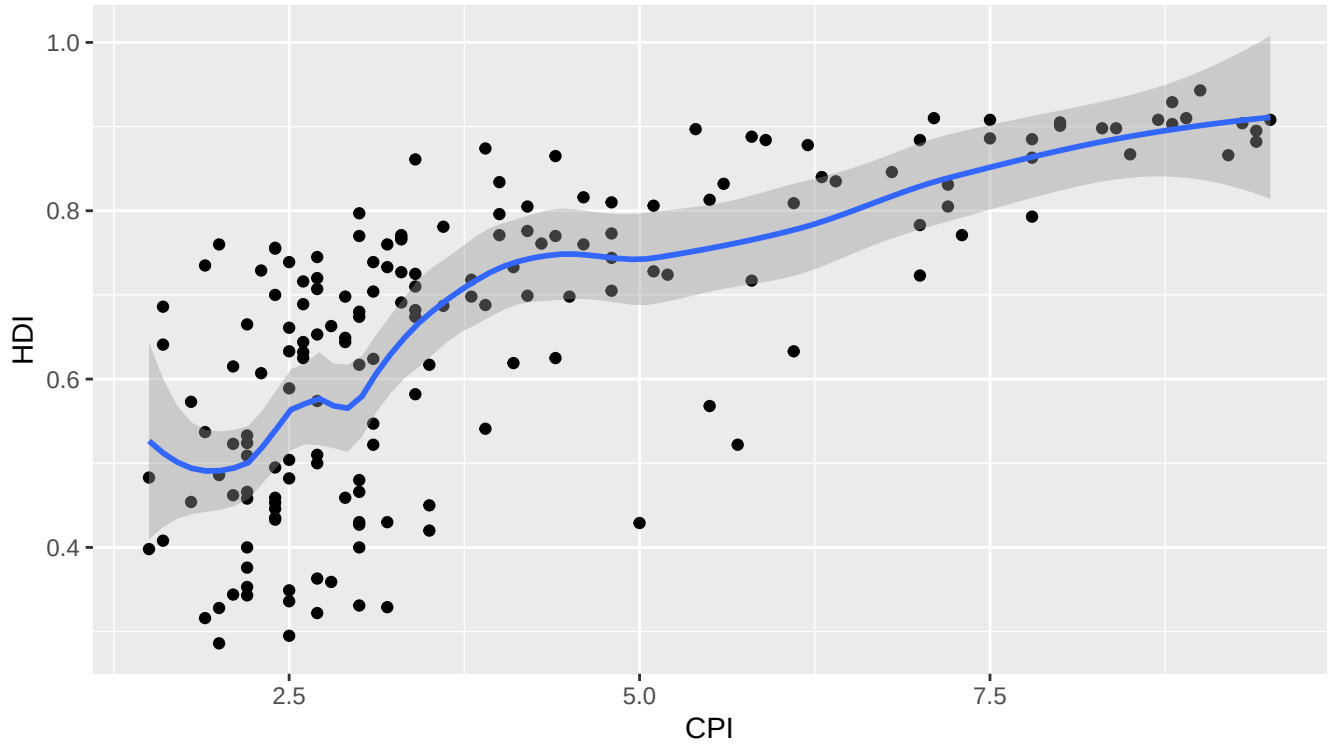
```
puntos <- p + geom_point()
puntos + geom_smooth(method = 'lm')
```




```
puntos + geom_smooth()
```



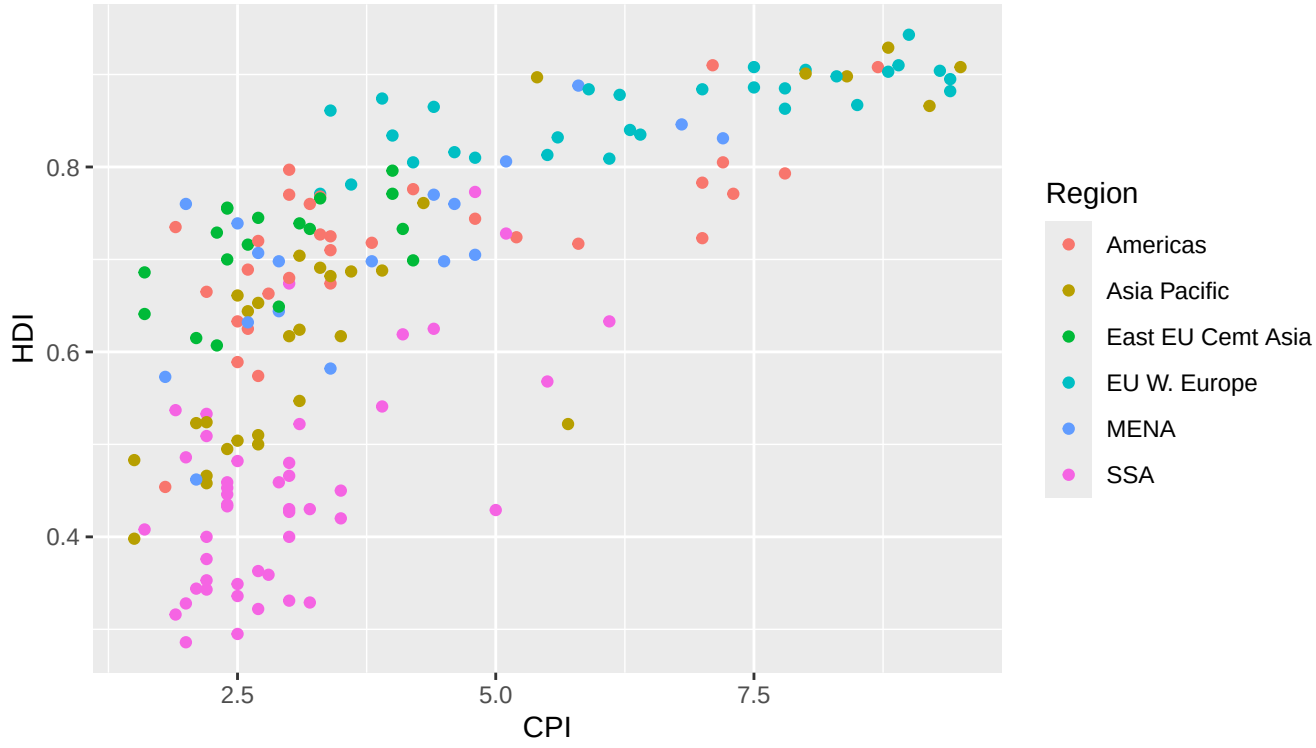

```
puntos + geom_smooth(method = 'loess', span = 0.4)
```



4. Etiquetas y colores

Se nombraron los ejes y la leyenda y se eligieron colores para las regiones.

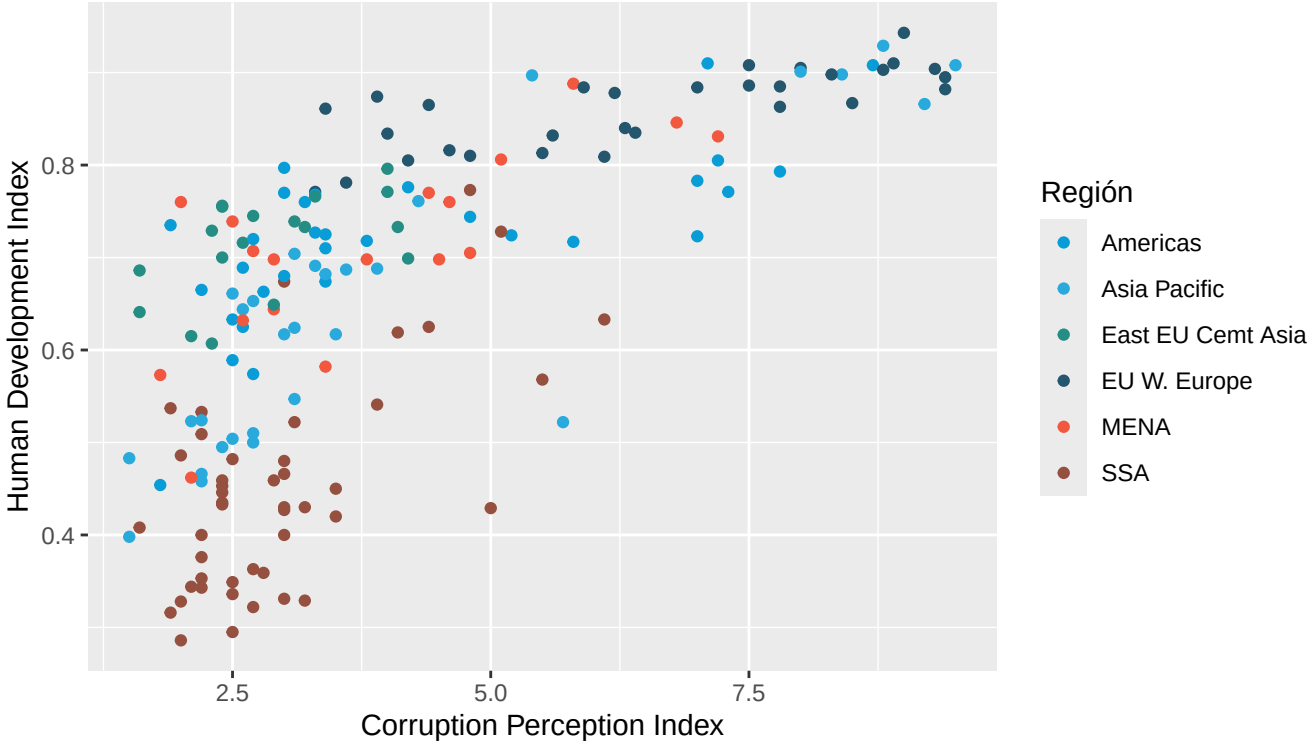
```
colores <- c('EU W. Europe' = '#24576D', 'Americas' = '#099DD7',
             'Asia Pacific' = '#28AADC', 'East EU Cemt Asia' = '#248E84',
             'MENA' = '#F2583F', 'SSA' = '#96503F')
regiones <- ggplot(dat, aes(x = CPI, y = HDI, colour = Region)) + geom_point()
regiones
```




```
etiquetado <- regiones + labs(x = 'Corruption Perception Index',
                              y = 'Human Development Index',
                              colour = 'Región')
etiquetado
```




```
etiquetado + scale_colour_manual(values = colores)
```

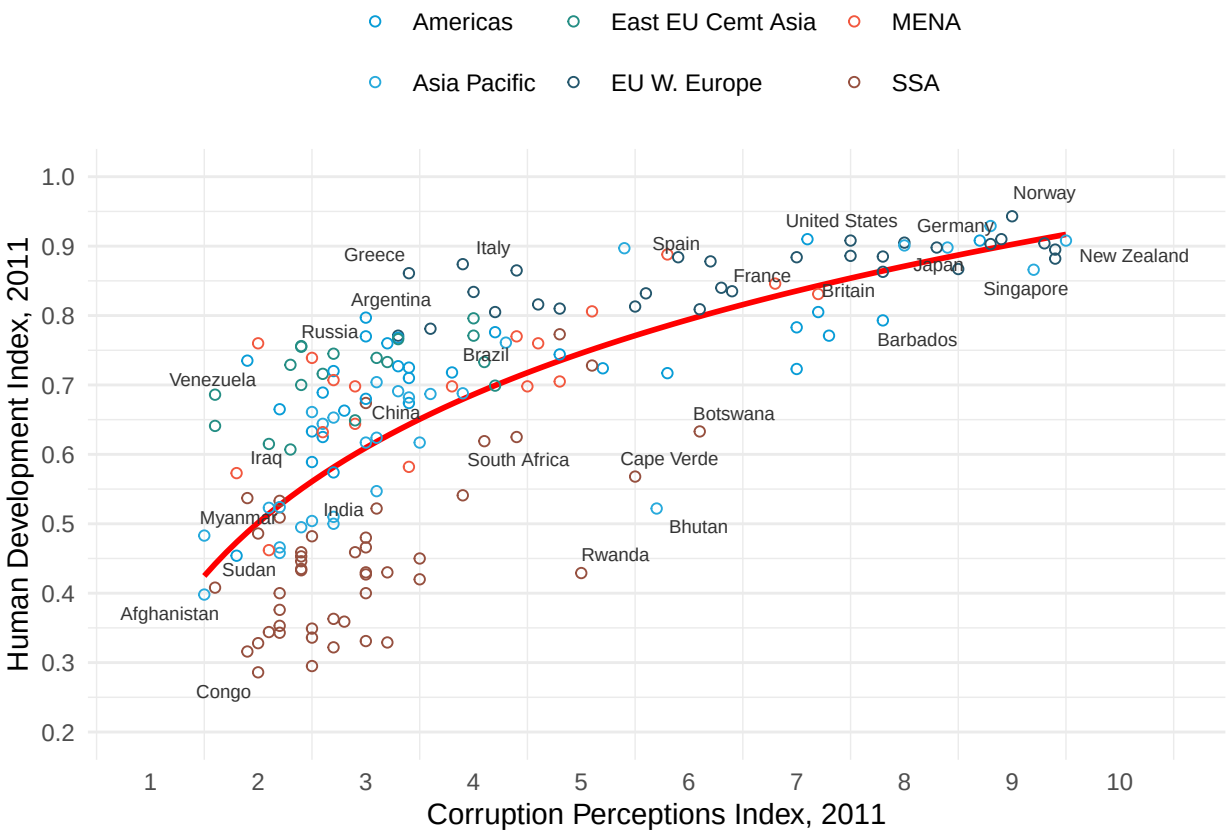


5. Gráfico de The Economist

Se añadieron el ajuste logarítmico, etiquetas de países, fuentes y R^2.

```
paises <- c('Russia', 'Venezuela', 'Iraq', 'Myanmar', 'Sudan',
            'Afghanistan', 'Congo', 'Greece', 'Argentina', 'Brazil',
            'India', 'Italy', 'China', 'South Africa', 'Spain', 'Botswana',
            'Cape Verde', 'Bhutan', 'Rwanda', 'France', 'United States',
            'Germany', 'Britain', 'Barbados', 'Norway', 'Japan',
            'New Zealand', 'Singapore')
modelo <- lm(HDI ~ log(CPI), data = dat)
ggplot(dat, aes(x = CPI, y = HDI, colour = Region)) +
  geom_smooth(aes(group = 1), method = 'lm', formula = y ~ log(x),
              se = FALSE, colour = 'red') +
  geom_point(shape = 1) +
  ggrepel::geom_text_repel(data = subset(dat, Country %in% paises),
                           aes(label = Country), colour = 'grey20',
                           size = 2.5, seed = 1, max.overlaps = Inf,
                           show.legend = FALSE) +
  scale_colour_manual(values = colores) +
  scale_x_continuous(limits = c(0.9, 10.5), breaks = 1:10) +
  scale_y_continuous(limits = c(0.2, 1), breaks = seq(0.2, 1, 0.1)) +
  labs(title = 'Corruption and Human development',
       x = 'Corruption Perceptions Index, 2011',
       y = 'Human Development Index, 2011', colour = NULL,
       caption = paste0('Sources: Transparency International; UN HDR. R2 = ',
                        round(summary(modelo)$r.squared, 2))) +
  guides(colour = guide_legend(nrow = 2)) +
  theme_minimal() + theme(legend.position = 'top', panel.grid.major.x = element_blank())
```

Corruption and Human development



Sources: Transparency International; UN HDR. R2 = 0.52